Navigation-Grade Accelerometer Calibration & Characterization — Equipment Reference

Centrifuge · Rate Table / Indexer · Optical Metrology | Specifications, Vendors & Key Considerations

IEEE 836-2009 / IEEE 1293-2018
NIST Traceable · ITAR/EAR Applicable

WHAT EACH INSTRUMENT CHARACTERIZES

Centrifuge

- **K2** — scale factor nonlinearity (quadratic term)
- **Asymmetry** — K1+ vs K1− at positive and negative g
- **Ki** — anisoelastic sensitivity (a_p·a_a cross-product term)
- **Cross-axis response** at elevated acceleration levels
- Full-range model validation beyond the ±1g gravity limit

Note: Bias (K0) and scale factor (K1) are most accurately determined by static multiposition tumble per IEEE 836 — not centrifuge.

Rate Table / Indexer

- **K0** — bias, the zero-g output offset
- **K1** — scale factor gain accuracy
- **Mip, Mia** — input axis misalignment (pendulous and output axes)
- **Non-orthogonality** — cross-axis sensitivity between sensor axes
- **Thermal coefficients** of all above when paired with a thermal chamber

Method: 6-position minimum or 24-point tumble test. A 2-axis table covers all 3 accelerometer axes in a single setup without repositioning.

Optical Metrology

- **Theodolite:** Table rotation axis vs. local gravity; fixture-to-table angular offset; optical cube face orthogonality verification
- **Autocollimator:** Per-position platen wobble detection; automated high-resolution orientation correction at each index position
- Decouples *fixture misalignment* from *intrinsic sensor IA misalignment*
- Anchors the NIST-traceable orientation reference chain for all calibration data

Priority: Theodolite alone covers setup characterization and fixture/sensor decoupling and is the recommended starting point before adding autocollimator.

CENTRIFUGE — KEY SPECIFICATIONS

Parameter	Target Spec	Notes
Acceleration Range	≥ 2× sensor full scale	*Must exceed sensor FS to characterize nonlinearity at extremes (e.g. ±50g for a ±20g sensor)*
Angular Velocity Stability	≤ 5 ppm	*ω ripple maps directly to acceleration error: a = ω²r; dominant uncertainty source for K2*
Radius Knowledge	≤ 50 ppm	*δa/a = δr/r + 2δω/ω; thermal expansion of arm must be managed — Invar arms or active temperature monitoring preferred*
Sub-table Index Accuracy	≤ 1 arcmin (arc-sec preferred)	*Positions DUT input axis relative to centripetal acceleration vector; IA misalignment mixes K1 into K2 fit*
Dynamic Balance	Verified	*Arm imbalance introduces cross-axis vibration corrupting sensor data; automated static/dynamic balance preferred*
Standards Compliance	IEEE 836-2009	*MIL-STD-810 if environmental survivability testing also required on the same platform*

RATE TABLE / INDEXER — KEY SPECIFICATIONS

Parameter	Target Spec	Notes
Angular Position Accuracy	≤ 5 arc-sec (1–2 preferred)	*Sets the floor on K0/K1 characterization. Request accuracy including wobble and runout — encoder resolution alone is a much more favorable number*
Platen Wobble / Runout	Specified separately from encoder	*Rotating platen tilt introduces a time-varying apparent misalignment that the encoder cannot detect*
Number of Axes	2-axis minimum	*2-axis: all 3 accelerometer axes in one setup; 3-axis: adds arbitrary motion profiles and simultaneous gyro calibration*
Drive Type	Direct-drive brushless + optical encoder	*Eliminates gear lash, backlash, and stiction that corrupt arc-second level position accuracy*
Thermal Chamber	Integrated preferred	*Required for full K0(T) and K1(T) characterization across operating temperature range*
Rotation Mode	Slip ring or wire wrap	*Slip ring: unlimited continuous rotation; wire wrap: lower electrical noise for quasi-static tumble*

KEY PROCUREMENT & INTEGRATION CONSIDERATIONS

Before You Specify

- **Error budget first** — define allowable calibration uncertainty for each model term before writing equipment specs; tester accuracy should be 1/3–1/5 of sensor allowable error per term (test uncertainty ratio)
- **Encoder accuracy vs. resolution** — always request accuracy including wobble and runout, not just encoder resolution; the two numbers differ significantly
- **Centrifuge arm expansion** — confirm thermal expansion management; Invar arms preferred for sub-100 ppm radius stability across operating temperature
- **Table axis orthogonality** — errors in the table's own axes appear directly as apparent sensor misalignment in multi-axis tumble data reduction

Facility & Infrastructure

- **Vibration isolation** — both instruments require isolated foundations; centrifuge may need a dedicated safety exclusion zone per facility safety requirements
- **Thermal stability** — ±0.5°C or better for lab environment; air column thermal gradients between autocollimator and optical cube introduce beam steering errors that appear as angular measurement noise
- **Lab sightlines** — plan theodolite and autocollimator positions before finalizing floor layout; install permanent survey monuments (tooling balls or kinematic mounts) for repeatable instrument repositioning
- **Cable pass-through** — confirm slip ring current rating and cable thermal rating for integrated thermal chamber operation

Compliance & Traceability

- **ITAR/EAR** — navigation-grade test equipment is export-controlled; foreign national access and any hardware export require early compliance review and appropriate licensing
- **NIST traceability** — centrifuge angular velocity and radius, table angular position, and all temperature sensors must have documented calibration chains traceable to NIST
- **Annual calibration** — budget instrument downtime and recalibration cost; theodolite verified via 360° closure checks; autocollimator via NIST-traceable optical polygon
- **Governing standards** — IEEE 836-2009 (centrifuge test procedures), IEEE 1293-2018 (accelerometer test procedures and model equation specification)

Equipment Recommendations & Optical Metrology

Vendor Examples · Rate Tables · Centrifuges · Optical Metrology · Theodolite & Autocollimator Integration

EQUIPMENT RECOMMENDATIONS

Rate Tables / Indexers

Model	Axes	Accuracy	Notes
Ideal Aerosmith 2002P/2002PG	2	±1 arc-sec	*Top-tier nav-grade; 18–36" platen; slip ring or wire wrap; thermal chamber option; ATL control language; direct-drive brushless motors with precision encoders*
Ideal Aerosmith 1542C-12-TL	2	±8 arc-sec	*Mid-tier; unlimited rotation on both axes; compact 12" platen; strong price/performance for mid nav-grade programs; same ATL control language*
Exail EVO-20 Series	2	Nav-grade class	*Full range from MEMS production to nav-grade characterization; modular; nGine controller; ProaXe GUI; thermal chamber integration option*
Acutronic 2/3-Axis Series	2–3	Customer spec	*World leader in precision motion simulators; highly configurable; strong defense/aerospace pedigree; custom accuracy tiers available*

Centrifuges

Model	g Range	Notes
Exail CTR Series	Customer spec	*IEEE 836-2009 compliant; brushless motors + optical encoders + slip rings; optional climatic chamber, vibration shaker, satellite sub-table; automated static/dynamic balance*
Ideal Aerosmith Centrifuge Series	Up to 1,000g / ≤200 lb	*Broad g range; same ATL control language as Ideal Aerosmith rate tables — simplifies cross-instrument test automation integration; sub-table indexer available*

Optical Metrology

Instrument	Vendors / Examples	Cost
Geodetic Theodolite	*Leica, Trimble, Sokkia — 1 arc-sec class*	$15–30K
Electronic Autocollimator	*Möller-Wedel ELCOMAT 3000, Taylor Hobson, Micro-Radian — 0.1 arc-sec*	$15–40K
Precision Optical Cube	*Zygo, Äpre Instruments — specify face orthogonality < 2 arc-sec*	$1–5K
Laser Tracker (alt.)	*Leica AT series, FARO — replaces theodolite + some autocollimator functions*	$100K+

OPTICAL METROLOGY — ROLE & INTEGRATION

Theodolite — Setup & Periodic Survey

- **Table axis vs. gravity** — establishes rotation axis tilt relative to local vertical; this systematic error aliases uniformly into every tumble position and must be corrected in data reduction
- **Fixture-to-table angular offset** — measures the offset between the optical cube reference frame and the table mechanical frame at the reference index position; recorded as a known correction to separate fixture misalignment from sensor IA misalignment
- **Cube face orthogonality** — verifies manufacturer orthogonality specification under actual lab thermal conditions before relying on the cube as an angular reference
- **Periodic re-survey** — after table maintenance, fixture rebuild, or significant lab thermal event; repositioned to permanent survey monuments for inter-campaign repeatability
- **Recommended starting point** — theodolite alone provides the orientation anchor needed to decouple fixture from sensor IA misalignment without autocollimator investment

Autocollimator — Per-Position Runtime Monitoring

- **Fixed externally in lab frame** — reads the optical cube riding on the fixture; independent of table encoder, providing a second measurement channel for orientation truth
- **Per-position angle correction** — digital angle output logged at each index position and substituted for nominal encoder angles in the misalignment fit; significantly tightens fit residuals on tables with 10–30 arc-sec encoder accuracy
- **Platen wobble detection** — catches dynamic tilt of the rotating platen that the table encoder cannot observe; important for tables with modest bearing specifications
- **Test automation integration** — electronic autocollimators output RS-232 or USB digital angle data; integrates into automated tumble test sequences without operator intervention at each position
- **Distinction from theodolite** — narrow field, differential, high-resolution (±10–30 arcmin range); the theodolite establishes macro geometry once; the autocollimator monitors it at high resolution during every test run

CALIBRATION & CHARACTERIZATION WORKFLOW SUMMARY

Phase 1 — Lab Setup & Geometric Reference

- **Level rate table** using precision electronic tilt sensors (0.1 arc-sec class)
- **Mount optical cube** on fixture co-located with sensor; bond to sensor case reference surface with rigid, thermally stable adhesive
- **Theodolite survey** — establish table rotation axis tilt vs. local vertical; measure cube face orientation relative to table axis; verify cube orthogonality. Record all offsets as fixed corrections for data reduction
- **Install permanent survey monuments** so theodolite can be repositioned consistently between campaigns

Phase 2 — Static Calibration (Rate Table)

- **6-position or 24-point tumble** — extract K0, K1, Mip, Mia via least-squares fit to gravity projections at known orientations
- **Temperature sweep** — repeat tumble at ≥ 5–10 temperature setpoints; extract K0T, K1T; allow ≥ 30–60 min soak per setpoint; confirm ±0.1°C chamber stability before collection
- **Thermal hysteresis test** — full up/down thermal cycle; record output at T_ref pre- and post-cycle to separate hysteresis from K0T
- **Allan Variance** — stationary DUT at stable temperature; extract bias instability and ARW from log-log plot

Phase 3 — Dynamic Characterization (Centrifuge)

- **Index DUT** using sub-table to place input axis radially; verify alignment with theodolite or autocollimator before spin-up
- **Multi-g sweep** — run centrifuge at multiple ω settings covering full sensor range and beyond; extract K2 and asymmetry from quadratic fit to residuals after removing K0 and K1
- **Multi-orientation high-g runs** — reindex DUT to multiple orientations to separate Ki·a_p·a_a (anisoelastic) from K2·a_in² (nonlinearity) geometrically; IA radial vs. tangential is the key orientation pair
- **Temperature-soaked centrifuge runs** — for K2 vs. temperature characterization; allow full thermal equilibration before each run; sequential runs preferred over active ramp

Sensor Error Model & Error Budget

IEEE 1293-2018 Standard Model · Term Definitions · Instrument Mapping · Nav-Grade Performance Bands

IEEE 1293-2018
Q-Flex · VBA · Nav-MEMS

IEEE 1293 STANDARD ACCELEROMETER MODEL EQUATION

STANDARD FORM — LINEAR + SECOND-ORDER TERMS · TECHNOLOGY-AGNOSTIC · APPLIES TO Q-FLEX, VBA, AND NAV-MEMS SENSORS · A_{IN} = INPUT AXIS ACCEL; A_P, A_A = PENDULOUS/OUTPUT AXIS ACCELS

$$a_{out} = K0 + K1 \cdot a_{in} + K2 \cdot a_{in}^2 + Mip \cdot a_p + Mia \cdot a_a + Ki \cdot a_p \cdot a_a + \varepsilon$$

MODEL TERM DEFINITIONS

Term	Name	Units	Description
K0	Bias	µg	*Zero-g output offset; drifts with temperature, time, and power cycles*
K1	Scale Factor	ppm	*Input-to-output gain; ppm error equals % full-scale output error*
K2	Nonlinearity	µg/g²	*Quadratic deviation from ideal linear input-output response*
Mip	Cross-Axis (Pendulous)	µrad	*Sensitivity to acceleration along the pendulous axis; ideally zero*
Mia	Cross-Axis (Output)	µrad	*Sensitivity to acceleration along the output axis; ideally zero*
Ki	Anisoelastic Coef.	µg/g²	*Cross-axis g²-sensitivity; product of pendulous and output axis accelerations (a_p·a_a); characterized on centrifuge by varying cross-axis loading geometrically*
ε	Residual / Noise	—	*Velocity random walk, quantization noise, vibration rectification error*

DATA REDUCTION — KEY RELATIONSHIPS & NOTES

Tumble Test (Rate Table)

- **K0 extraction:** $K0 = ½(V_{+g} + V_{-g})$ — average of ±g pair; scale factor cancels exactly
- **K1 extraction:** $K1 = ½(V_{+g} - V_{-g}) / g$ — difference normalized to gravity reference
- **Mip, Mia:** Cross-axis gravity projections at known orientations excite misalignment terms; extracted via least-squares fit across full orientation set
- **Use corrected orientation angles** — substitute theodolite-measured offsets for nominal encoder angles as gravity projection reference; residuals tighten significantly

Centrifuge

- **Centripetal acceleration:** $a = \omega^2 \cdot r$ | $\delta a/a = 2 \cdot \delta\omega/\omega + \delta r/r$ — ω stability is the dominant uncertainty term
- **K2 extraction:** Quadratic fit to $(V - K0 - K1 \cdot a)$ vs. a^2; residual computed relative to the tumble-derived linear model — decouple K0 and K1 first
- **Ki separation:** Run DUT with IA radial vs. tangential; $Ki \cdot a_p \cdot a_a$ appears only under simultaneous cross-axis loading — geometrically separable from $K2 \cdot a_in^2$ by orientation selection

Key Principles

- **Thermal decouple:** Run full ±g pair at each temperature setpoint to extract K0(T) and K1(T) independently; fitting both from a single-axis measurement confounds the two
- **Test uncertainty ratio:** Tester accuracy ≤ 1/3 to 1/5 of sensor allowable error per term; document TUR for each parameter in the calibration record
- **Ellipsoid fitting:** For multi-axis IMU calibration, nonlinear least-squares fit to a 3D ellipsoid simultaneously recovers bias vector, scale factor matrix, and misalignment matrix from multi-position tumble data

TERM → PRIMARY INSTRUMENT MAPPING

Model Term	Primary Instrument	Test Method	Optical Metrology Role
K0 — Bias	*Rate Table + Theodolite*	*6- or 24-point tumble; average of ±g orientations eliminates scale factor*	*Theodolite verifies table axis tilt to gravity — tilt error aliases uniformly into K0 at every position*
K1 — Scale Factor	*Rate Table / Centrifuge*	*Difference of ±g tumble positions; centrifuge for high-g determination*	*Axis tilt shifts effective gravity projection, biasing K1; centrifuge radius accuracy dominates at high g*
K2 — Nonlinearity	*Centrifuge*	*Multiple ω settings across full g-range; quadratic fit to residuals after removing K0 and K1*	*Sub-table index accuracy critical — IA misalignment mixes K1 into K2 fit; index error sets K2 floor*
Mip, Mia — Misalignment	*Rate Table + Theodolite*	*Multi-axis tumble; cross-axis gravity projections excite misalignment terms in the model fit*	*Theodolite critical to separate fixture misalignment from intrinsic sensor IA misalignment*
Ki — Anisoelastic	*Centrifuge*	*Multiple DUT orientations varying a_p·a_a cross-loading; Ki·a_p·a_a separable from K2·a_in² by geometry — IA radial vs. tangential orientations isolate each term*	*Sub-table index accuracy determines ability to isolate Ki from K2 — both are g²-class terms reported in µg/g²*
ε — Noise / ARW	*Rate Table (static)*	*Allan Variance on stationary DUT at constant temperature; vibration shaker for VRE characterization*	*Not applicable — noise is a statistical characterization, not geometric*

TIERED ERROR BUDGET — NAVIGATION-GRADE PERFORMANCE BANDS

☐ **High-End Nav-Grade** — Quartz pendulous (Q-Flex), Vibrating Beam Accelerometer (VBA); bias <10 µg class ☐ **Mid Nav-Grade** — Precision MEMS, quartz-MEMS hybrid; bias 10–100 µg class

Model Term	Units	High-End Nav	Mid Nav	Implied Tester Requirement	Primary Instrument
K0 — Bias	µg	< 10 µg	10–100 µg	*Table position ≤ 1–2 arc-sec; axis tilt to gravity < 5 arc-sec (theodolite verified)*	*Rate Table + Theodolite*
K1 — Scale Factor	ppm	< 5 ppm	5–50 ppm	*Table position < 1 arc-sec for tumble; centrifuge radius ≤ 50 ppm for high-g determination*	*Rate Table / Centrifuge*
K2 — Nonlinearity	µg/g²	< 10 µg/g²	10–100 µg/g²	*Centrifuge ω stability ≤ 5 ppm; radius knowledge ≤ 50 ppm; sub-table index ≤ 1 arcmin*	*Centrifuge*
Mip, Mia — Misalignment	µrad	< 10 µrad	10–100 µrad	*Table axis orthogonality < 5 arc-sec; theodolite for fixture/sensor decoupling; platen wobble specified separately*	*Rate Table + Theodolite*
Ki — Anisoelastic	µg/g²	< 1 µg/g²	1–10 µg/g²	*Centrifuge with precision sub-table; IA radial vs. tangential orientations separate Ki·a_p·a_a from K2·a_in² — sub-table index accuracy is the limiting factor*	*Centrifuge*
Bias Instability / ARW	µg / µg/√Hz	< 1 / < 1	1–10 / 1–10	*Static rate table; Allan Variance analysis; vibration shaker for VRE characterization; thermally stable and vibration-quiet environment*	*Rate Table (static)*

Thermal Error Modeling & Characterization

IEEE 1293-2018 Thermal Extensions · Term Definitions · Test Method Summary · Thermal Chamber Specifications & Thermal Error Budget

IEEE 1293 THERMAL MODEL EXTENSION

BASE COEFFICIENTS EXPANDED AS POLYNOMIAL FUNCTIONS OF TEMPERATURE — FIRST AND SECOND ORDER SHOWN; ALL TERMS RELATIVE TO T_REF (CAL. REFERENCE TEMPERATURE)

$K0(T) = K0_0 + K0T \cdot \Delta T + K0T2 \cdot \Delta T^2$ [bias thermal model; ΔT = T − T_ref]

$K1(T) = K1_0 + K1T \cdot \Delta T + K1T2 \cdot \Delta T^2$ [scale factor thermal model; mismatch between T_ref and operating temperature compounds every other thermal error term]

THERMAL TERM DEFINITIONS

Term	Name	Units	Definition & Practical Notes
K0T	Bias Temp. Coef.	µg/°C	*Linear rate of change of zero-g output with temperature. Dominant thermal error for most nav-grade sensors. Extracted from steady-state soak sweep. Second-order term K0T2 (µg/°C²) captures curvature; needed for ranges > 60°C span.*
K1T	Scale Factor Temp. Coef.	ppm/°C	*Linear rate of change of gain with temperature. Must be extracted separately from K0T — data reduction decouples the two at each setpoint. Second-order K1T2 (ppm/°C²) significant for ranges > 80°C.*
Hys	Thermal Hysteresis	µg	*Residual output difference returning to T_ref after a full thermal excursion. Not captured by K0T. Often the largest practical thermal error; caused by stress relief and creep in mounting structures.*
S_trans	Transient Sensitivity	µg/(°C/min)	*Output error during active temperature ramp, not at steady state. Caused by thermal gradients across sensor body. Requires controlled ramp-rate test; distinct from K0T. Critical for airborne and munition applications.*
TOR	Turn-On Repeatability	µg (1σ)	*Bias variation across repeated cold-start power cycles at the same temperature. Distinct from in-run bias stability. Driven by stress relaxation and electronics settle. Requires n ≥ 10 cycles.*
T_ref	Cal. Reference Temp.	°C	*Temperature at which $K0_0$ and $K1_0$ are defined. All thermal terms are relative to this point. A mismatch between T_ref and operating temperature compounds every other thermal error. Document explicitly; typically +20°C or +25°C.*

THERMAL TEST METHOD SUMMARY

Term(s)	Method	Key Requirement
K0T, K1T, K0T2, K1T2	*Stepped soak sweep — rate table tumble at each setpoint*	*≥ 5–10 setpoints across range; ≥ 30–60 min soak per setpoint; ±0.1°C stability confirmed before data collection*
Thermal Hysteresis (Hys)	*Full up/down thermal cycle to T_min and T_max, returning to T_ref*	*Soak at each extreme ≥ 2× sensor thermal time constant before reversing; record output at T_ref pre- and post-cycle to isolate from K0T*
Transient Sensitivity (S_trans)	*Controlled ramp-rate test — sensor operating during active temperature ramp*	*Chamber ramp rate programmable and repeatable (1–5°C/min typical); record output continuously during ramp; normalize error to ramp rate*
Turn-On Repeatability (TOR)	*Repeated cold-start power cycles at T_min; full cool-down between each*	*n ≥ 10 power cycles; record bias at thermal equilibrium each cycle; report 1σ variation across sample*
K2 Temp. Sensitivity	*Centrifuge runs at multiple temperature-soaked conditions*	*Full thermal equilibrium required before each centrifuge run; sequential runs preferred over simultaneous centrifuge + active ramp*

THERMAL ERROR BUDGET

Term	Units	High-End Nav	Mid Nav
K0T	µg/°C	< 1	1–10
K1T	ppm/°C	< 1	1–10
Hys	µg	< 5	5–50
S_trans	µg/(°C/min)	< 1	1–10
TOR (1σ)	µg	< 5	5–50
K2 vs. Temp	µg/g²/°C	< 0.5	0.5–5

THERMAL CHAMBER — KEY SPECIFICATIONS

Required for Nav-Grade Thermal Characterization

- **Temperature Range:** −55°C to +85°C (mil-range); −40°C to +70°C (most commercial nav-grade)
- **Setpoint Stability:** ±0.1°C or better — stability, not just accuracy; data collection begins only after stability is confirmed
- **Ramp Rate Control:** Programmable 0.5–5°C/min; max rate alone is insufficient — low-rate controllability required for transient sensitivity testing
- **Spatial Uniformity:** ≤ 0.5°C across DUT footprint; gradients drive apparent transient sensitivity even at steady state
- **Rate Table Integration:** Cable pass-through port; confirm cable thermal rating; optical window if autocollimator monitoring during thermal test
- **Soak Time:** ≥ 3–5× sensor thermal time constant; quartz pendulous sensors may require 20–40 min per setpoint
- **NIST Traceability:** Chamber temperature sensor on traceable calibration chain; uncertainty documented in test record

Pitfall — Hysteresis vs. K0T: If output at T_ref after a thermal cycle does not match the pre-cycle value, the discrepancy is hysteresis — not temperature coefficient. Always record output at T_ref before and after each excursion to separate the two effects.